

Unbiased exact diagonalization of quantum spin ice in the π -flux regime: removing finite-size loops by translationally-invariant twist averaging

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Abstract

The thermodynamics and dynamics of quantum spin ice (QSI) are governed by an emergent compact $U(1)$ gauge field. Its unfrustrated 0-*flux* sector ($J_{\pm} > 0$) is accessible to sign-problem-free quantum Monte Carlo (QMC); the frustrated π -*flux* sector ($J_{\pm} < 0$) realized in the dipolar-octupolar cerium pyrochlores carries a QMC sign problem, leaving exact diagonalization (ED) as the only unbiased finite-temperature method. ED has a finite-size problem specific to QSI: a periodic cluster hosts short ice-preserving ring exchanges that wind the torus and have no infinite-lattice analogue. The dominant ones are four-site loops with coupling $g_4 = 4J_{\pm}^2/J_{zz}$, which is *second* order in $J_{\pm}$ and hence parametrically larger than the genuine photon coupling $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$; they contaminate exactly the emergent-gauge observables (the low- T specific-heat peak and the low-energy $S^{zz}(\mathbf{q}, \omega)$). We remove them by twist averaging in the translationally-invariant gauge: threading a $U(1)$ flux through the cluster as a uniform vector potential on *every* bond and averaging the effective Hamiltonian over the flux. The winding loops offset the magnetization (S^z) pattern across the cluster, so the flux dephases them, while genuine bulk ring exchanges – balanced, with no such offset – are untouched. The average works, but only if three conditions the naive procedure violates are met: the gauge must be uniform (the boundary-only gauge removes merely half), the average must span the doubled period set by the half-integer S^z dipole (the naive 2π leaves a 40% residual), and it must act on the Hamiltonian rather than on an observable. On the 16- and 32-site clusters this moves the low- T gauge peak and the low-energy S^{zz} weight from the artifact scale g_4 to the photon scale g_{hex} for both flux signs, while leaving the high- T charge (spinon) peak – genuine, uncontaminated physics – untouched. This makes a 32-site pyrochlore an unbiased laboratory for the π -flux gauge sector, to be benchmarked against 0-flux QMC and applied at the exchange parameters of the cerium pyrochlores.

1 The problem

Quantum spin ice and its emergent photon. The minimal model of QSI is the XXZ antiferromagnet on the pyrochlore lattice, in the local $\langle 111 \rangle$ frames,

$$H = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+), \quad 0 < |J_{\pm}| \ll J_{zz}. \quad (1)$$

For $J_{\pm} \rightarrow 0$ the ground manifold is the exponentially degenerate set of 2-in-2-out (ice-rule) states. A small $J_{\pm}$ lifts the degeneracy through virtual ring exchanges and generates the deconfined $U(1)$

lattice gauge theory of QSI: S^z is the emergent electric field, the third-order hexagonal ring exchange

$$g_{\text{hex}} = \frac{12 |J_{\pm}|^3}{J_{zz}^2} \quad (2)$$

is the Maxwell term whose quantum is the emergent photon, and ice-rule defects are gapped, fractionalized spinons (pair threshold $\Delta_s \approx 0.8 J_{zz}$ on the clusters below). The *sign* of the ring coupling follows $\text{sgn}(J_{\pm}^3)$: $J_{\pm} > 0$ gives the unfrustrated 0-flux vacuum, $J_{\pm} < 0$ the frustrated π -flux vacuum. Thermodynamically the photon appears as a low- T Schottky peak in $C(T)$ at $T \sim g_{\text{hex}}$, far below the spinon (charge) peak at $T \sim J_{zz}$; dynamically it appears as low-energy weight in the non-spin-flip channel $S^{zz}(\mathbf{q}, \omega)$.

Why π -flux needs exact diagonalization. In the S^z basis the off-diagonal matrix elements of eq. (1) are $-J_{\pm}$. For $J_{\pm} > 0$ they are negative and worldline/stochastic-series QMC is sign-problem-free: the 0-flux QSI is quantitatively characterized. For $J_{\pm} < 0$ they are positive and QMC acquires a sign problem. The π -flux regime – believed to describe $\text{Ce}_2\text{Zr}_2\text{O}_7$, $\text{Ce}_2\text{Sn}_2\text{O}_7$, $\text{Ce}_2\text{Hf}_2\text{O}_7$ – is thus a QMC dead zone, and unbiased finite- T information must come from exact diagonalization of a finite cluster.

The finite-size problem. A periodic cluster is a flat torus, and its periodic identifications generate ice-preserving ring exchanges that *wind* the torus with no counterpart on the infinite lattice. The shortest and most damaging are *four-site* loops. Because a four-site loop factorizes into two disjoint nearest-neighbour (NN) hops, it is generated at *second* order in $J_{\pm}$,

$$g_4 = \frac{4 J_{\pm}^2}{J_{zz}}, \quad \text{so} \quad \frac{g_4}{g_{\text{hex}}} = \frac{J_{zz}}{3|J_{\pm}|} \gg 1. \quad (3)$$

The artifact is thus *parametrically larger* than the physics, and grows as $|J_{\pm}| \rightarrow 0$ – the weak-coupling regime of the cerium materials. It lives entirely in the ice manifold, so it contaminates precisely the emergent-gauge observables: it pushes the low- T $C(T)$ peak and the low-energy S^{zz} weight up to the scale g_4 instead of g_{hex} (quantified in section 4). The spinon continuum (S^{+-} at $\omega \sim J_{zz}$) lies outside the ice manifold and is *not* contaminated.

Goal. Remove this finite-size contamination from the gauge sector of a periodic-cluster ED, so that the 32-site pyrochlore delivers unbiased π -flux thermodynamics and dynamical structure factor.

2 The effective gauge Hamiltonian and the winding loops

Cluster torus. The cluster carries N sites with periodic identifications generated by the rows of an integer matrix $\mathbf{L} \in \mathbb{Z}^{3 \times 3}$ (in units of the cubic lattice constant a); position space is the flat torus $T^3 = \mathbb{R}^3 / \mathbf{L}\mathbb{Z}^3$ with first homology $\mathbb{Z}^3$. Each NN bond (i, j) has a *wrap vector* $\mathbf{n}_{ij} = \text{round}[(\mathbf{r}_j - \mathbf{r}_i)\mathbf{L}^{-1}]$, zero in the interior and nonzero across the boundary.

Effective Hamiltonian. Degenerate $(J_{\pm})$ -perturbation theory in the ice manifold yields $H_{\text{eff}} = \sum_{\mathcal{C}} c_{\mathcal{C}} O_{\mathcal{C}}$, a sum of ring-exchange operators $O_{\mathcal{C}} = (S_a^+ S_b^- S_c^+ S_d^- \cdots)$ over ice-preserving cycles $\mathcal{C}$; because each intermediate resolvent is Ising ($\propto J_{zz}$), $c_{\mathcal{C}} \propto J_{\pm}^{|\mathcal{C}|/2}$ exactly. On the $2 \times 2 \times 2$ FCC cluster ($N = 32$) the ice manifold has dimension 2970, so H_{eff} is *fully diagonalizable* and its thermodynamics is numerically exact. Enumeration (Schrieffer–Wolff with per-process bookkeeping) gives table 1: the winding four-loops and wrapping hexagons are the contamination; the contractible hexagons carry the photon.

Table 1: Census of ice-preserving ring exchanges. “Winding/wrapping” loops close only through the periodic boundary ($\mathbf{w} \neq \mathbf{0}$); “contractible” hexagons are genuine bulk plaquettes ($\mathbf{w} = \mathbf{0}$). The last column is the survival fraction under the twist average of section 3.

cluster	channel	count	order	kept by the average
$1 \times 1 \times 1$ cubic ($N=16$)	winding four-loops	36	$J_{\pm}^2$	0
	wrapping hexagons	48	$J_{\pm}^3$	0
	contractible hexagons	16	$J_{\pm}^3$	1
$2 \times 2 \times 2$ FCC ($N=32$)	winding four-loops	24	$J_{\pm}^2$	0
	wrapping hexagons	96	$J_{\pm}^3$	0
	contractible hexagons	32	$J_{\pm}^3$	1

3 The method: twist averaging in the translationally-invariant gauge

The label that separates artifact from physics. The winding loops share one feature that no genuine bulk ring exchange has: they offset the magnetization (S^z) pattern *across* the cluster. Define the S^z dipole of a ring exchange O_C (which raises a site set R , lowers a set L) as

$$\delta_C = \rho_C + \mathbf{N} \cdot \mathbf{L}, \quad \rho_C = \sum_{i \in R} \mathbf{r}_i - \sum_{j \in L} \mathbf{r}_j, \quad (4)$$

the net spatial displacement of the flipped S^z : ρ is the intrinsic dipole (home positions), $\mathbf{N} \cdot \mathbf{L}$ the boundary-wrap contribution. (This is the microscopic S^z associated with S_{tot}^z conservation – to which the twist flux below couples – *not* the emergent electric charge; the ice manifold is the charge-free vacuum, and O_C creates no spinons. The net S_{tot}^z is conserved, so δ measures whether the process’s S^z current *winds* the torus.) A contractible hexagon closes in real space and is balanced, $\delta = \mathbf{0}$; a winding four-loop closes only through the boundary, so its flipped spins cannot balance and its S^z pattern is offset by half a cell, $\delta \neq \mathbf{0}$ (fig. 1). A twist flux dephases exactly the loops with $\delta \neq \mathbf{0}$.

The average. Thread a $U(1)$ flux φ through the cluster as a uniform vector potential $\mathbf{A} = \mathbf{L}^{-1}\varphi$ on *every* bond ($\theta_{ij} = \mathbf{A} \cdot \mathbf{d}_{ij}$, $\mathbf{d}_{ij}$ the minimum-image displacement), so that each ring exchange picks up the phase of its S^z dipole, $c_C \rightarrow c_C e^{-i\mathbf{A} \cdot \delta_C}$, and average the *Hamiltonian* over the flux:

$$\overline{H_{\text{eff}}} = \int H_{\text{eff}}(\varphi) \frac{d\varphi}{(\text{period})^3}, \quad c_C \rightarrow c_C \overline{e^{-i\varphi \cdot (\mathbf{L}^{-1}\delta_C)}} = c_C [\delta_C \in \mathbf{L}\mathbb{Z}^3]. \quad (5)$$

The physical loop ($\delta = \mathbf{0}$) is kept; the winding loop ($\delta \neq \mathbf{0}$) is dephased away, at every order in $J_{\pm}$. This is the familiar twist-averaging idea, and it works – but only if three conditions the naive procedure violates are met:

- (i) *Uniform gauge.* The phase must sit on every bond. If the flux sits only on the boundary bonds, a winding loop has a virtual path through interior bonds alone that carries no phase and survives, leaving *half* the coupling.
- (ii) *Doubled period.* A winding loop offsets its S^z pattern by half a lattice vector ($\mathbf{L}^{-1}\rho = \frac{1}{2}$; verified for all winding four-loops and wrapping hexagons on both clusters), so its phase

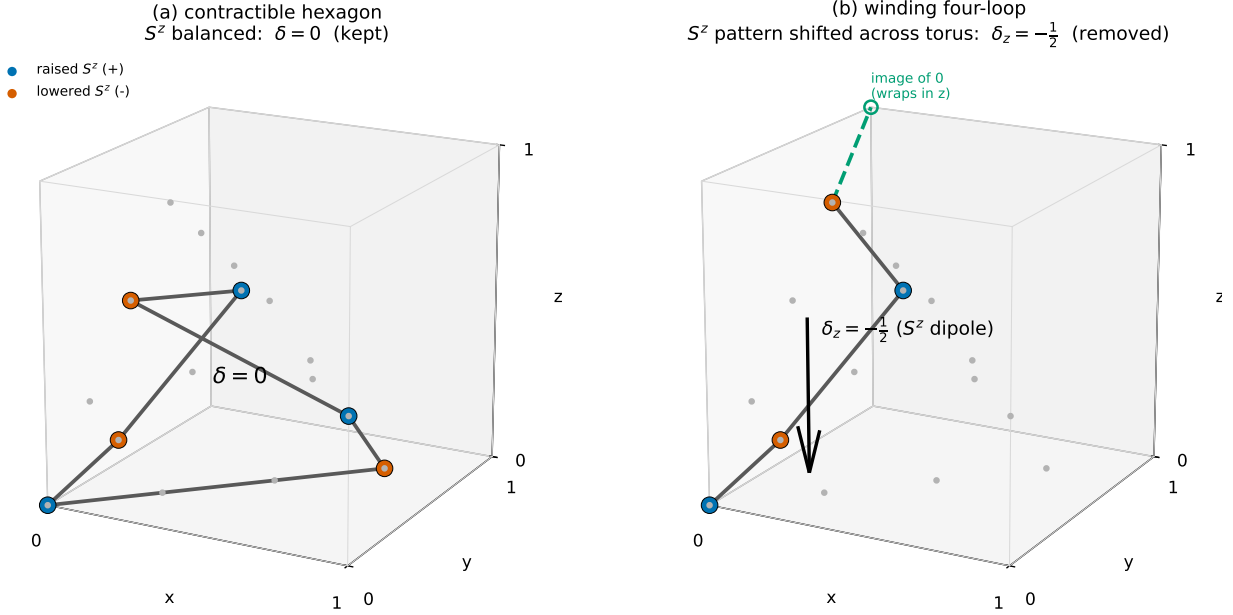


Figure 1: The S^z dipole of a ring exchange on the 16-site cubic pyrochlore cell, $\delta = \sum_{\text{raised}} \mathbf{r} - \sum_{\text{lowered}} \mathbf{r}$. (a) A contractible hexagon (sites 0, 2, 7, 4, 14, 15) is balanced – raised (+) and lowered (–) sites share a centroid – so $\delta = \mathbf{0}$: physics. (b) A torus-winding four-loop (0, 2, 7, 5) closes only through the periodic boundary (dashed bond to the periodic image of site 0, wrapping in z); its S^z pattern is offset by half a cell, $\delta_z = -\frac{1}{2}$: a finite-size artifact. (δ is the microscopic magnetization dipole, not the emergent spinon charge, which is absent in the ice manifold.)

$e^{-i\varphi/2}$ is periodic only modulo 4π . Averaging over the naive $[0, 2\pi)^3$ leaves a $\sim 40\%$ residual; the period $[0, 4\pi)^3$ is required.

- (iii) *Operator level.* One must average H_{eff} , then diagonalize. Averaging a specific observable $C(T)$ or $S(\mathbf{q}, \omega)$ instead keeps the spurious scale in any gauge.

4 Demonstration

The average gives the winding four-loop coefficient (and the resulting low- T specific-heat peak) shown here, at $J_{\pm} = -0.1 J_{zz}$ on the 16-site cluster, isolating each of the three conditions:

scheme	winding coeff. / g_4	$T_{\text{peak}}/g_{\text{hex}}$
none (bare)	−1.00	3.18
boundary gauge (violates (i))	−0.50	1.63
smooth gauge, $[0, 2\pi)^3$ (violates (ii))	+0.40	0.84
smooth gauge, $[0, 4\pi)^3$ (correct)	0.00	0.62

Only the uniform gauge over the doubled period removes the loop (0/0/1 survival of table 1, to machine precision, with the contractible hexagon kept at $+g_{\text{hex}}$). Figure 2 shows the full $C(T)$ from exact 16-site diagonalization (12870 states): the cleaned low- T gauge (photon) peak sits at the physical scale g_{hex} , while the high- T charge (spinon) peak at $T \sim J_{zz}/3$ – genuine physics outside the ice manifold – is untouched, as it must be. On the 32-site FCC cluster (exact full

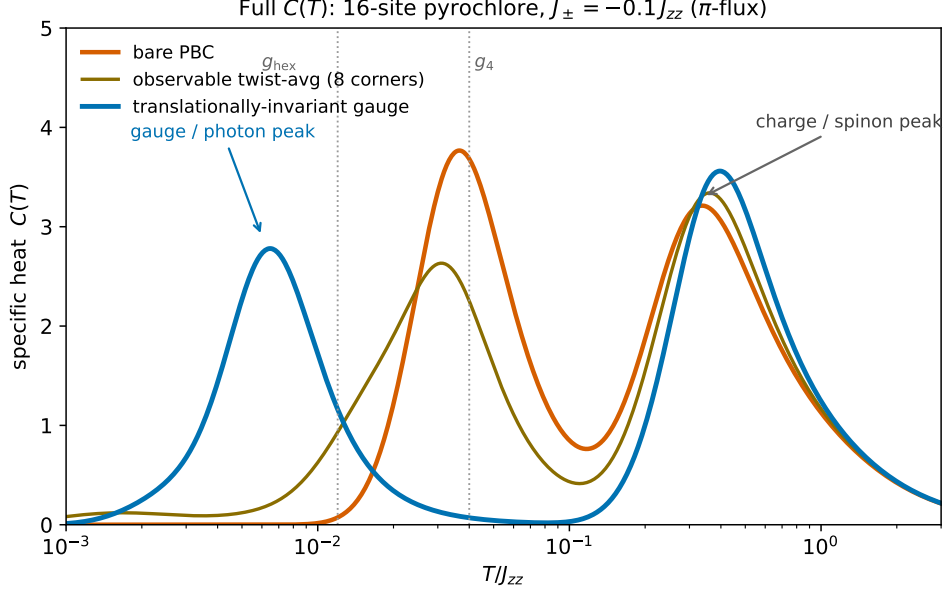


Figure 2: Full specific heat of the 16-site π -flux cluster ($J_{\pm} = -0.1 J_{zz}$). Bare periodic ED (orange) places the low- T gauge peak at the finite-size artifact scale g_4 ; averaging the *observable* over eight corners (olive) barely moves it; translationally-invariant twist averaging of the *Hamiltonian* (blue) moves it to the photon scale g_{hex} . The high- T charge (spinon) peak at $T \sim J_{zz}/3$ is genuine, uncontaminated, and unchanged.

spectrum of the 2970-state ice manifold) the bare gauge peak sits at the artifact scale and drifts up as $g_4/g_{\text{hex}} = J_{zz}/3|J_{\pm}|$ toward weak coupling, while the average holds it at $O(g_{\text{hex}})$; the same holds for the low- ω weight of $S^{zz}(\mathbf{q}, \omega)$ on both clusters and both flux signs.